⚙️ RATCHET IMPROVEMENT REPORT · March 15, 2026

Before & After

Autonomous code improvement run on @ratchet-run/cli

BEFORE		AFTER
83	→ 0 points	83

Issues: **683** → **675** (8 fixed)

5	4	1	22m 23s
Clicks run	Landed clean	Rolled back	Total time

📋 Ratchet ran 5 clicks in 22m 23s, fixing 8 issues. 1 click was rolled back due to test failures.

Issues Fixed

All issue types detected — counts before and after the run

🏆 **Top 1 Win**

↓ **14** console.log calls in src

ISSUE	SEVERITY	BEFORE	AFTER	CHANGE
console.log calls in src	low	60	**46**	**−14**
repeated code lines	medium	333	**333**	—
functions >50 lines	medium	27	**27**	—
any types	medium	4	**4**	—
await-in-loop patterns	medium	4	**4**	—
missing auth/security controls	medium	1	**1**	—
dead code indicators (TODO, commented code)	low	8	**8**	—
lines >120 chars	low	246	**252**	**+6**

Score Breakdown

Category-by-category improvement

CATEGORY	BEFORE	AFTER	MAX	CHANGE
🧪 Testing	23	23	25	—
🔒 Security	13	13	15	—
📝 Type Safety	15	15	15	—
⚠️ Error Handling	16	16	20	—
⚡ Performance	7	7	10	—
📖 Code Quality	9	9	15	—

before / after

What Ratchet Did

Every change made — 5 clicks, each tested before committing

✅ Click 1 Landed

Fixed issues across 1 file

commands/torque.ts`

✅ Click 2 Landed

Fixed issues across 3 files

lib/cli.ts` — added `validateProjectEnv(cwd)` that consolidates `warnIfStaleBinary` + `assertIsRepo` + `warnIfDirtyWorktree` + `loadConfigOrExit` into a single reusable function, commands/torque.ts` — removed local `STATE_FILE` constant, replaced 4-line startup block with `await validateProjectEnv(cwd)`, imports `STATE_FILE` from `status.ts`, commands/improve.ts` — same: removed local `STATE_FILE`, replaced 4-line startup block with `await validateProjectEnv(cwd)`, imports `STATE_FILE` from `status.ts`

↩️ Click 3 Rolled back

Fixed issues across 4 files

core/improve-report-html.ts, commands/improve.ts, commands/torque.ts, lib/cli.ts

✅ Click 4 Landed

Fixed Remove or replace console.log calls with the structured logger. For debug-only logs, remove entirely. For operational logs, use logger.info/debug. [ROI: 0.08, +4pt next tier, +4pt max] across 0 files

✅ Click 5 Landed

Fixed Remove or replace console.log calls with the structured logger. For debug-only logs, remove entirely. For operational logs, use logger.info/debug. [ROI: 0.08, +4pt next tier, +4pt max] across 1 file

commands/simulate.ts

User Impact Assessment

Persona simulation — how real users would experience these changes

OVERALL SENTIMENT

Users generally completed the flow with neutral-to-positive sentiment, but pre-flight check UX created consistent friction through jargon-heavy warnings and missing recovery guidance.

Top Pain Points

• Dirty worktree warning uses git jargon with no explanation and no clear call-to-action (mentioned by 2/3 personas: Marcus, Sofia)

• Stale binary warning is opaque jargon with no inline fix command (mentioned by 2/3 personas: Marcus, Sofia)

• No visual or contextual separation between the pre-flight validation phase and the main run output, causing confusion about what belongs to which phase (Marcus)

• Pre-flight checks run silently with no progress indication, making it hard to identify which check failed (Sofia)

• No labeling of warnings as pre-flight/setup issues vs. user errors or bugs, causing misattribution (Sofia)

Top Suggestions

• Rewrite the dirty worktree warning in plain language and include an inline suggested fix: 'You have uncommitted changes. Run `git stash` to proceed cleanly, or pass --allow-dirty to continue.' (Marcus + Sofia)

• Replace 'stale binary' with plain language and always include the fix command inline: 'Your ratchet install may be outdated — run `npm install -g @ratchet-run/cli` to update.' (Marcus + Sofia)

• Add a visible pre-flight checklist (e.g. '✓ Git repo detected ✓ Config loaded ✗ Uncommitted changes found') so users see all checks and their status at once (Sofia)

• Add a clear visual divider (blank line + header or dashes) between the pre-flight validation summary and the click loop output (Marcus)

• On rollback, show the last 5 lines of test output inline so users don't need to hunt for the cause (e.g. 'Tests failed: 2 assertions in src/api.test.ts') (Marcus)

Persona Results

Priya Kapoor (power-user) — neutral

Marcus Chen (casual) — positive

Sofia Rodriguez (new-user) — neutral